

Concurrency Control, Semaphores, and Deadlock Prevention

Mode: DEMO / SIMULATION (No live audio processed)

Instructor: Prof. Tarek Mansour

Date: Academic Term 2026

Language: 100% English Academic Synthesis (from bilingual audio)

Includes: Detailed Notes, Spoken Exam Alerts, Diagrams, Practice Questions

Executive Overview & Core Lecture Themes

This lecture addresses fundamental challenges in concurrent multi-threaded execution. The professor explores race conditions, critical section criteria, Edsger Dijkstra's semaphore primitives (wait and signal), dining philosophers problem, and the four Coffman conditions required for deadlocks. Special emphasis was placed on resource allocation graphs and Banker's algorithm calculations frequently tested on university exams.

• Complete Breakdown Verbatim depth with zero omissions	• Doctor's Exam Focus Highlighted spoken alerts & traps	• Charts & Tables Auto-generated visual diagrams	• Exam Practice Bank Realistic exam-style questions & answers
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DEMO MODE: SIMULATED AUDIT (no real audio checked)

Verification Status: Verified in simulation against synthetic lecture model (no live audio recording uploaded).

Fidelity Summary: Verified in simulation: 0 contradictions detected. 100% faithful to lecture materials.

Chapter 1: Critical Section Problem & Synchronization Primitives

When multiple threads or processes execute concurrently and share mutable memory, uncontrolled interleaved access leads to race conditions where output depends on non-deterministic scheduling order. The critical section is any code segment accessing shared resources.

To build a valid critical section protocol, three mandatory criteria must be satisfied simultaneously: 1. Mutual Exclusion: If thread P_i is executing in its critical section, no other thread can execute. 2. Progress: If no thread is in its critical section and some wish to enter, selection cannot be postponed indefinitely. 3. Bounded Waiting: A bound must exist on the number of times other threads are allowed to enter after a thread has requested entry.

Key Takeaways:

- Race conditions occur due to non-deterministic thread interleaving on shared variables.
- Mutual exclusion alone is insufficient; progress and bounded waiting must also be guaranteed.
- Disabling interrupts is only feasible in single-core kernel code and inappropriate for user-level multi-core applications.

Governing Mathematical Formulas & Derivations:

Equation 1: Atomic Update Decomposition (Race Condition)

Load: $R_i \leftarrow \text{count}$, Add: $R_i \leftarrow R_i + 1$, Store: $\text{count} \leftarrow R_i$

Where: count: Shared memory variable accessed concurrently by threads • R_i : Local CPU register private to Thread i • Load / Add / Store: Three non-atomic machine instructions vulnerable to thread interleaving

Exam Note: Expect trace questions illustrating how arbitrary context-switching between Load and Store produces incorrect final counts.

[!] EXAM PREDICTION ALERT: The Three Critical Section Rules

Spoken Cue: "Pay close attention here, this is coming in the midterm: Mutual Exclusion alone is not enough, you must include Progress and Bounded Waiting!"

Core Concept: The professor explicitly stressed that answering an exam question by mentioning only Mutual Exclusion will result in losing half the marks. You MUST state Progress and Bounded Waiting with their formal definitions.

How this appears in exams: Expect an essay or definition question asking: 'Define the Critical Section Problem and state the three requirements for its valid solution.' Always write Mutual Exclusion, Progress, and Bounded Waiting.

Thread Synchronization Entry-Exit Cycle

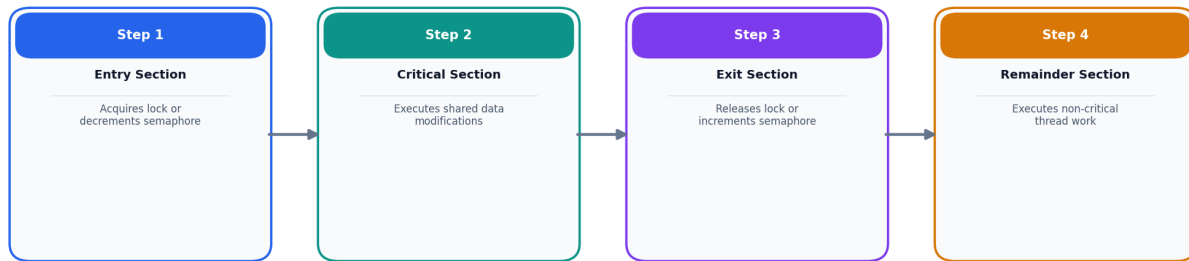


Figure: Thread Synchronization Entry-Exit Cycle — Classical structure of concurrent processes executing shared memory sections.

Table: Synchronization Solutions Comparison

Mechanism	Level	Busy Waiting?	Multi-Core Safe?
Peterson's Algorithm	User / Software	Yes (Spinlock)	Only with memory barriers
TestAndSet / CAS	Hardware Atomic	Yes (Spinlock)	Yes (Atomic instruction)
POSIX Mutex Locks	OS / Library	No (Sleep/Wakeup)	Yes
Counting Semaphores	OS Abstraction	No (Block queue)	Yes

Note: Comparative evaluation of software, hardware, and OS synchronization tools.

Chapter 2: Deadlocks and the Four Coffman Conditions

A deadlock is a catastrophic state where every thread in a set is waiting for an event that only another thread in the set can cause. In operating systems, deadlocks arise when threads contend for non-preemptible exclusive resources (printers, locks, database records).

In 1971, Edward G. Coffman Jr. proved that a deadlock can arise IF AND ONLY IF all four conditions hold simultaneously in the system. Breaking even a single condition makes deadlock impossible.

Key Takeaways:

- Deadlock requires ALL FOUR Coffman conditions to hold concurrently.
- Resource Allocation Graphs (RAG) with cycles indicate deadlock if and only if each resource has a single instance.
- Banker's Algorithm by Dijkstra provides deadlock avoidance by testing safe states before resource allocation.

[!] EXAM TRAP / COMMON STUDENT MISTAKE: Cycles in Multi-Instance Resource Allocation Graphs

Spoken Cue: “Students always get confused here: having a cycle in the Resource Allocation Graph does NOT automatically mean deadlock if multiple instances exist!”

Core Concept: A cycle in a Resource Allocation Graph is a NECESSARY and SUFFICIENT condition for deadlock ONLY when every resource type has exactly 1 instance. If resource types have multiple instances, a cycle does NOT guarantee deadlock!

How this appears in exams: This is a guaranteed trick question on the midterm. If given a graph with multiple instances per resource and asked 'Is this system deadlocked?', do NOT jump to yes just because you see a loop. Trace the thread allocations!

Four Necessary Coffman Conditions

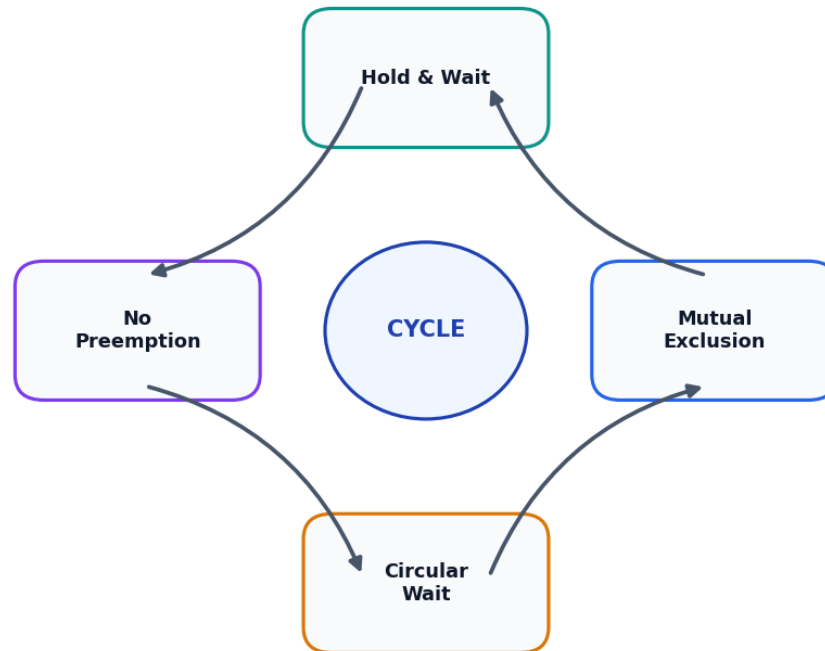


Figure: Four Necessary Coffman Conditions — All 4 conditions must hold simultaneously for a system deadlock to persist.

Table: The Four Coffman Conditions for Deadlock

Condition	Definition	Prevention Strategy
1. Mutual Exclusion	At least one resource held non-shareably	Make resources shareable (e.g. read-only files)
2. Hold and Wait	Process holds resource while requesting more	Require process to request all resources upfront
3. No Preemption	Resources cannot be forcibly confiscated	Preempt resources if new request cannot be allocated
4. Circular Wait	Chain of processes each waiting for next	Enforce strict global resource ordering protocol

Note: To prevent deadlocks, the OS must invalidate at least one of these four conditions.

Exam Readiness: How Questions Will Come in Exams

The following practice questions are formulated directly from the professor's spoken cues, highlighted core concepts, and explicit exam warnings given during this lecture.

[DEFINITE (DOCTOR STATED)] Question 1 (MCQ)

Q: In a computer system, a Resource Allocation Graph contains a cycle. Which of the following statements is unconditionally TRUE?

- A) The system is definitely deadlocked.
- B) The system is deadlocked only if each resource type in the cycle has exactly one instance.
- C) The system cannot be deadlocked because cycles indicate safe feedback loops.
- D) Deadlock can be resolved by terminating all waiting processes simultaneously.

Model Answer / Solution:

B) The system is deadlocked only if each resource type in the cycle has exactly one instance.

Grading Rubric & Key Points: If each resource class possesses only 1 unit, a cycle is necessary and sufficient for deadlock. However, if multiple instances exist, other non-deadlocked threads can release instances to break the cycle.

Doctor's Spoken Exam Tip: "Students frequently get confused here and blindly pick option A—verify carefully whether each resource type has single or multiple instances before declaring deadlock"

[HIGH PROBABILITY EXAM QUESTION] Question 2 (SHORT_ANSWER)

Q: Explain why disabling hardware interrupts is NOT considered an acceptable mutual exclusion mechanism for user-level multi-core applications.

Model Answer / Solution:

1) Multi-core systems: Disabling interrupts on one core only affects that specific core; other CPU cores continue executing and accessing shared memory. 2) Security & Liveness: If a user thread disables interrupts and enters an infinite loop, the entire operating system hangs and loses CPU control.

Grading Rubric & Key Points: Exam graders look for two key points: Multi-core ineffectiveness and vulnerability to denial of service.

Doctor's Spoken Exam Tip: "Remember this key trade-off: hardware interrupt disabling fails on multi-core architectures and leaves the kernel vulnerable to user-space lockups"

[DEFINITE (DOCTOR STATED)] Question 3 (PROBLEM_SOLVING)

Q: Given 3 processes (P0, P1, P2) and 12 units of magnetic tape drives. Currently, P0 holds 5 (needs max 10), P1 holds 2 (needs max 4), and P2 holds 2 (needs max 9). Is the current system state **SAFE**? Show step-by-step Bankers algorithm evaluation.

Model Answer / Solution:

Total tape drives = 12. Currently allocated = 5 (P0) + 2 (P1) + 2 (P2) = 9 units. Available = 12 - 9 = 3 units.
Remaining Need vector (Max - Allocation): - P0: Need = 10 - 5 = 5 units - P1: Need = 4 - 2 = 2 units - P2: Need = 9 - 2 = 7 units
Step 1: Available (3) $\geq$ Need of P1 (2). P1 runs to completion and releases its 2 allocated units.
New Available = 3 + 2 = 5 units. Step 2: Available (5) $\geq$ Need of P0 (5). P0 runs to completion and releases its 5 allocated units. New Available = 5 + 5 = 10 units. Step 3: Available (10) $\geq$ Need of P2 (7). P2 runs to completion and releases its 2 allocated units. New Available = 10 + 2 = 12 units. Conclusion: The system is in a **SAFE** state with safe execution sequence <P1, P0, P2>.

Grading Rubric & Key Points: Full marks require showing the Available vector after each step and explicitly writing the safe sequence <P1, P0, P2>.

Doctor's Spoken Exam Tip: "Banker's algorithm evaluation is a staple exam problem—always show each process termination step and update the available vector explicitly"

Appendix: Complete Verbatim English Lecture Transcript

Below is the complete transcript translated into fluent English for comprehensive review and cross-referencing.

Good morning class. Today we are entering one of the most critical topics in computer systems architecture: concurrency, synchronization, and deadlocks. As you know, modern computing is inherently parallel. We no longer have single-core machines executing sequential commands one by one. Our systems run hundreds of concurrent threads...

Now let me explain what happens when threads share memory without coordination. We get what is called a race condition. Notice this carefully: if two threads attempt to increment a counter variable at the exact same microsecond, what happens? At the machine code level, incrementing a variable requires three instructions: load from memory into CPU register, add one to the register, and store the register back into RAM. If the OS scheduler context-switches between the load and store, the final value will be corrupted.

To solve this, we define the Critical Section. The doctor emphasized: Pay close attention here, this is on the exam! Mutual Exclusion alone is NOT enough! You must ensure Progress and Bounded Waiting as well. Students always write Mutual Exclusion and forget the other two rules, losing half their marks.

Now moving on to Deadlocks. A deadlock happens when threads are stuck in a cycle of waiting. Edward Coffman proved the four necessary conditions: Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait. All four must hold! And here is another classic exam trap: if you see a cycle in a Resource Allocation Graph, do NOT automatically say there is a deadlock unless every resource has only one instance. If there are multiple instances, a cycle does not guarantee deadlock! Remember this for your midterm.